

2. What is Interaction Design / HCI really about?

The whole course is basically about **how people and computers talk to each other** – not in code, but through buttons, screens, gestures, sound, etc.

Examples on early slides:

- A **Game Boy / Wii U style controller** – physical buttons, shapes.
- Kids playing **motion-sensing games** with their bodies (like Kinect) – using gestures instead of pressing buttons.

These are all *interactive systems*.

Different names for similar ideas (Slide 5)

You'll hear many terms:

- **User-centered design (UCD)**
 - Focuses on *the user and their tasks*.
 - Example: designing a shopping website based on how real users search, add to cart, and pay.
- **Human-centered design (HCD)**
 - A bit broader: cares about a person's **cognitive, physical, and emotional** needs.
 - Not just “user of app”, but full human life context.
- **People-centered design**
 - Very similar to HCD but more casual – emphasizes *real people* in everyday life.
- **Customer experience design (CX)**
 - More business-focused. Looks at the **entire journey** with a company: ads → website → app → customer support.

👉 **MEMORIZE:** at least these four terms and one-line meanings.

3. Good vs Bad Design – how HCI sees the world

This whole section is just “spot the stupidity” and “spot the cleverness” in designs.

Good Design: TiVo remote (Slide 6)

Why is this remote praised?

- **Peanut shape** – fits naturally into your hand.
- **Logical layout** – buttons grouped by function.
- **Color-coded / distinct buttons** – easy to find important ones by feel and sight.
- **You barely need to look at it** after some use.

👉 **MEMORIZE:** 3 reasons TiVo remote is well-designed (shape, layout, color-coded/easy to locate).

Bad Design 1: Elevator panel (Slide 7)

- Bottom row: labels and buttons **look the same**, so people accidentally press the label instead of the button.
- Top row: numbers themselves act as labels, so users make fewer mistakes.

Lesson: **Labels and controls must look different** and be clearly mapped.

👉 This shows a problem of **poor signifiers / visibility** (user can't tell what is pressable).

Bad Design 2: Vending machine (Slide 8)

- To use it, you must **press a button first** to activate the bill reader.

- But users' normal mental model: **insert money first, then select item.**
- It **breaks the common convention**, so people get confused and think it's broken.

👉 **HCI point:** When you break a strong habit/convention, you must redesign very carefully or give strong guidance.

Bad Design 3: Road sign (Slide 9)

- Shows a “do not enter” symbol **and** a right arrow together.
- User message: “Don’t go here... but also go here.”
- Completely ambiguous → driver confusion.

👉 HCI lesson: **Communication must be clear and unambiguous.**

Bad Design 4: Remote with identical triangles (Slide 10)

- Volume and channel buttons are both just up/down triangles.
- Only difference: tiny “V” vs “CH” text.
- Hard to distinguish quickly → accidental channel changes when you just wanted volume.

👉 Lesson: **Important controls must be visually distinct.** Don't make users squint.

Bad Design 5: Calendars from badusability.com (Slide 11)

Examples:

- January: Big Titanic graphic literally **covers dates.**
- February: Chaotic layout, not standard calendar grid.

- November: Weird labels, missing dates, confusing placement.
- December: Better but still minor issues.

👉 HCI lesson: **Decoration must not destroy function**. Primary task of a calendar = see dates clearly.

4. Information overload + Outdated technology thinking

Too much data – Passport example (Slide 12)

Pakistani passport shown with:

- Tons of data fields, both human-readable and machine-readable.
- Important info can get lost in all that text and codes.

👉 HCI lesson:

Show **only what is necessary** at a time, in **clear, organized** form. More info ≠ better.

Outdated model of the “home computer” (Slide 13)

- An old RAND picture: giant machine filling a room, with many knobs and screens.
- It was their “future home computer.”

Lesson:

- Our **current design ideas are also temporary**; they will look silly later.
 - HCI keeps evolving with technology and culture.
-

5. Health & Accessibility – color blindness example (Slides 14–18)

Multiple Ishihara plates (color-dot circles with numbers):

- Some people clearly see “7”, “2”, “45”, “74”.
- A color-blind person might not see the numbers at all.

👉 HCI point:

You **cannot assume everyone sees like you**.

If your design uses only color (e.g., red vs green text), some users may miss critical info.

6. Another good design: Marble answering machine (Slide 19)

How it works:

- Every time someone leaves a message, a **marble** drops into a tray.
- To listen: put a marble into a “playback” cup.
- To call back: put the same marble into another cup.

Why it's good:

- Uses **familiar physical behavior** (objects = messages).
 - One marble = one message → clear mapping.
 - Only single, simple actions – no menus, no screens.
 - Feels **playful and intuitive**.
-

7. Everyday benefits of good design (Slide 20)

Examples your slides give:

- Using an ATM

- Playing a game
- Copy-pasting in spreadsheets
- Shopping online
- Updating social media status

In all these, **good interaction design**:

- Hides the complexity.
 - Lets you focus on your goal (get cash / post story), **not** on “how do I use this interface?”.
-

8. Dilemma: how to interact with a Smart TV? (Slides 22–23)

You have multiple possible input methods:

- **Standard remote + grid keyboard**
 - You “peck” letters one-by-one. Very slow.
- **Touchpad remote with 2 rows of letters**
 - Faster than grid, but still can be fiddly.
- **Voice control** (into remote or smart speaker)
 - Fast, natural speech. Great for searching movies like “Action movies with Tom Cruise”.
 - But struggles in noisy rooms or with accents/unusual names.
- **Physical keyboard (like Minuum)**
 - Fast typing, but bulky and weird for a living room.

👉 HCI lesson:

There is often **no one “best” interaction**. It depends on context: distance from screen, user skill, noise level, etc.

9. Switching to Digital (Slide 24)

Many tasks moved from physical to digital:

- Paying **electricity/gas bills**
- Booking **rides**
- Using **maps**
- **Bus/plane tickets** with QR codes instead of paper

Benefits:

- Faster, easier, more convenient.
- Info stored for next time (saved cards, saved trips).
- No need to stand in long queues.
- Just scan a QR code at barriers.

Disadvantages:

- **Fraud and privacy risks.**
- Some people **don’t want to share personal data online.**
- Some users **prefer talking to real humans**, especially for important purchases.

👉 **MEMORIZE:** at least 2 benefits + 2 disadvantages of going digital.

10. “What to design?” – core questions (Slide 25)

Before designing anything, you must know:

1. **Who** are the users?
 - children vs adults, experts vs beginners, disabled vs non-disabled.
2. **What** activities are they doing?
 - searching, learning, buying, playing, etc.
3. **Where** will they use it?
 - at home, in noisy streets, while walking, in a car, etc.

Goal: **optimize the interaction** so it matches users' needs in that context (who/what/where).

👉 **MEMORIZE this who–what–where triad.**

11. Definitions of Interaction Design (Slide 26)

Two exam-style definitions:

1. **Sharp, Rogers & Preece (2019):**

Designing interactive products that support how people communicate and interact in their everyday and working lives.

2. **Winograd (1997):**

Design of spaces for human communication and interaction.

In easier words:

- **Interaction design** = planning *how* people and technology “talk” to each other: what appears on screen, what input is used, and how the system responds.

👉 **MEMORIZE BOTH DEFINITIONS** (wording roughly close), and remember “everyday & working lives” and “spaces for communication and interaction”.

12. Goals of Interaction Design (Slide 27)

Main goals:

- **Develop usable products** (not just technically correct).
- **Involve people** (users) in the design process.
- Think about:
 - What people are **good at / bad at**
 - What might help them in their current ways of doing things
 - What would give **quality experiences**, not just functionality
- Respect **privacy** when collecting data.
- **Listen** to people’s needs and use **people-centered techniques** (interviews, testing, etc.).

👉 **MEMORIZE:** “usable products + involve people + consider abilities + protect privacy + use people-centered techniques + iterate”.

13. Interaction Design’s relationship with other fields

13.1 Interaction design as umbrella term (Slide 28)

Other design words that sit under ID:

- User interface design
- Software design
- Web design

- Product design
- Experience design (UX)

All of these are parts of the same big thing: **designing computer-based systems for people.**

13.2 Academic disciplines contributing to ID (Slide 29–30)

From the diagram and notes:

- **Psychology / Cognitive science** – perception, memory, attention.
- **Social sciences (sociology, anthropology)** – how groups behave, culture.
- **Computer science / software engineering** – actually building the systems.
- **Engineering** – hardware, infrastructure.
- **Ergonomics / Human factors** – physical comfort, safety, body measurements.
- **Informatics** – how information is structured and used.

👉 **MEMORIZE: at least 4 names of contributing disciplines.**

13.3 Design practices (Slide 31)

These are more artistic/creative fields that influence how interfaces *look and feel*:

- Graphic design
- Product design
- Industrial design
- Artist-design
- Film industry (storyboarding, etc.)

13.4 Interdisciplinary fields doing ID (Slide 32)

These are already mixtures of tech + people:

- HCI itself
- Ubiquitous computing (computers everywhere, embedded)
- Human factors
- Cognitive engineering & cognitive ergonomics
- CSCW (Computer-Supported Cooperative Work)
- Information systems

👉 **MEMORIZE:** at least HCI, ubiquitous computing, CSCW as interdisciplinary fields.

13.5 Multidisciplinary teams (Slide 33)

- **Many backgrounds** = designers, programmers, psychologists, business people.
 - **Benefits:** more varied ideas, better solutions.
 - **Disadvantages:** communication problems (different jargon, priorities).
-

13.6 Interaction design in business (Slide 34)

Examples of famous consultancies:

- **Nielsen Norman Group** – usability & human-centered products.
- **Cooper** – goal-directed design.
- **IDEO** – design thinking, products/services/environments.

You don't need deep info, but you should know they exist and what they roughly do.

14. People-Centered Design & User Experience (Slides 35–37)

People-centered design (Slide 35)

Focuses on:

- How people **feel** about a product.
- Pleasure and satisfaction using it, holding it, opening it.
- Overall impression and experience – not just “it works”.

Don Norman's idea (paraphrased): It's not enough that products work and are usable; they should also add **joy, excitement, pleasure, fun, and beauty** to people's lives.

👉 **MEMORIZE:** people-centered design = care about emotional experience, not just functionality.

Defining user experience (Slide 36)

UX =

- How users **perceive** a product (e.g., smartwatch looks sleek or chunky).
- Their **emotional reaction** when using it (relaxed, anxious, excited).

Hassenzahl's model:

- **Pragmatic qualities** – Is it useful, practical, simple to achieve goals?
- **Hedonic qualities** – Is it stimulating, fun, meaningful, supporting well-being?

👉 **MEMORIZE:** pragmatic = goal/utility; hedonic = pleasure/stimulation.

iPod user experience success (Slide 37)

- From the start, Apple gave a **high-quality UX**:
 - Simple interface (click wheel).
 - Elegant design, cool branding.
 - Integrated with iTunes for easy music management.
 - Became a **fashion item**, not just a music player.

Lesson: **Experience + emotion** can make a product win even if others have similar technical specs.

15. Core characteristics of interaction design (Slides 38–39)

Core characteristics (Slide 38):

1. **Involve users throughout** the project.
2. **Clearly define usability and UX goals** at the start.
3. **Iterate** (design → prototype → test → improve, repeatedly).

Why? (Slide 39)

- Helps designers understand real needs, not assumptions.
- Recognize that different user groups vary (teens ≠ elderly).
- Avoid stereotypes (e.g., not all older people need huge fonts).
- Be aware of both **sensitivities** (privacy, dignity) and **capabilities**.

👉 **MEMORIZE: involve users, clear goals, iteration.**

16. Accessibility & Inclusiveness (Slides 40–43)

Definitions (Slide 40)

- **Accessibility:**
The extent to which a product is usable by people with **disabilities** (vision, hearing, etc.). Example: screen readers, VoiceOver.
- **Inclusiveness:**
Designing so the **widest possible range of people** can use the product – regardless of disability, age, education, income. Example: smartphones designed for everyone.

👉 **MEMORIZE BOTH DEFINITIONS.**

Understanding disability (Slide 41–42)

Key ideas:

- Disability can **change over time** (aging, recovery).
- Impact varies during the day or by environment (e.g., glare).

Types of disability:

1. **Sensory** – vision or hearing issues (blindness, color blindness, deafness).
2. **Physical** – difficulty moving body parts (after stroke, injury).
3. **Cognitive** – learning difficulties, memory loss, dementia.

Duration categories:

- **Permanent** – long-term condition (e.g., paralysed legs).
- **Temporary** – broken arm, short-term illness.
- **Situational** – environment makes you effectively disabled (e.g., very noisy room where you can't hear an audio alert).

👉 **MEMORIZE: 3 types (sensory, physical, cognitive) and 3 durations (permanent, temporary, situational).**

Being cool about disability (Slide 43)

- Example: stylish prosthetic covers; “wearing their wheels” instead of just “using a wheelchair”.
 - Modern design tries to make assistive tech **fashionable and empowering**, not ugly or stigmatizing.
-

17. Cultural differences (Slide 44)

Example:

- 5/21/2023 vs 21/5/2023 – which is the date?
- Different countries use different formats.

Questions raised:

- Which format should be used for **international services**? (Often ISO: 2023-05-21).
 - Why are some products (like smartphones) accepted worldwide, but **websites** sometimes need heavy localization?
 - Because language, colors, images, layout, and symbols can mean different things in different cultures.
-

18. Usability vs User Experience goals (Slides 45–47)

Usability goals (Slide 45)

A product should be:

1. **Effective** – users can successfully complete tasks.
2. **Efficient** – done with reasonable time and effort.
3. **Safe** – prevents serious errors and protects users.
4. **Have good utility** – has the right functions.
5. **Easy to learn** – new users can pick it up quickly.
6. **Easy to remember** – returning users don't have to re-learn everything.
7. **Enjoyable to use** – not painful/boring.

👉 **MEMORIZE THIS 7-POINT LIST. Very exam-able.**

Usability vs UX goals (Slide 46–47)

- **Usability goals** = more **objective/measureable** (time to complete task, error rate, etc.).
- **User Experience goals** = more **subjective/emotional** (fun, satisfying, frustrating, etc.).

There can be **trade-offs**:

- A banking app may be secure and safe but feel annoying (many steps).
- A game may be very fun but not “safe” in the way a banking app must be.

User experience goal examples (Slide 47):

- **Desirable**: satisfying, enjoyable, fun, engaging, motivating, surprising, rewarding, emotionally fulfilling.
- **Undesirable**: boring, frustrating, creepy, patronizing, intrusive, annoying.

You don't have to memorize every word, but know the idea and a few examples.

19. Design Principles (Slides 48–62)

These are like **rules-of-thumb** or **do's and don'ts**.

19.1 Visibility (Slides 49–51)

- Users should **see what controls exist** and **know what they do**.

Bad example: elevator where you must insert your room card but that slot isn't obvious.

Better design:

- Make card reader prominent.
- Add label / flashing light.
- Give auditory instructions.

Another example: automatic sensor taps / hand dryers that don't react when someone wears black clothes.

👉 **MEMORIZE: Visibility = making relevant parts & actions obvious.**

19.2 Feedback (Slide 52)

- System should tell user **what just happened**.

Types: sound, highlighting, animation, vibrations.

Example: click a screen button → it changes color / appears pressed + maybe makes a “click” sound.

👉 **MEMORIZE: Feedback = sending info back to user about action's result.**

19.3 Constraints (Slide 53)

- Limits the user so they **can't perform wrong actions**.

Examples:

- Physical: key only fits one way into the lock.
- Digital: “Save” option disabled (greyed out) when no changes.

👉 **MEMORIZE: Constraints help prevent errors by restricting actions.**

19.4 Mapping / Logical design (Slides 54–55)

- Mapping = **relationship between controls and their effects.**

Ambiguous example:

- PC back with multiple identical round ports -> user doesn't know where to plug mouse/keyboard.

Better design:

- Put icon right next to the port (**adjacent mapping**).
 - Use color-coding so user can match connector to port.
-

19.5 Consistency (Slides 56–59)

- **Internal consistency:** same app uses similar style & actions everywhere.
- **External consistency:** different apps/devices use similar conventions.

Benefits:

- Easier to learn and remember.

Problems:

- When too many commands begin with same letter (Save, Spell, Select, Style), you break keyboard consistency (Ctrl+S, Ctrl+Shift+L etc.).

- External inconsistency example:
 - Phone keypad: 1–2–3 on top row
 - Calculator keypad: 7–8–9 on top row
→ user must switch mental models.

👉 **MEMORIZE:** definition of internal vs external consistency + keypad example.

19.6 Affordances (Slides 60–62)

- **Affordance:** a property that suggests *how* something can be used.

Physical examples:

- Button affords pressing.
- Handle affords pulling/turning.

In interfaces:

- We talk more about **perceived affordances**:
 - We *learn* that a shaded rectangle with text behaves as a clickable button.
 - Icons, scrollbars etc. visually suggest certain actions.

Activity on slide 62 asks: if you're a novice, would you know what to do with these on-screen objects? If not, the perceived affordance is weak.

👉 **MEMORIZE:**

- Affordance = suggests how to use.
 - In GUIs we rely on **perceived affordances** (learned conventions).
-

20. Final key points (Slide 63)

You should be able to summarize:

- **Interaction design** = designing interactive products to support how people communicate & interact in everyday/work life.
 - Aim: **quality user experiences**.
 - It's **multidisciplinary** (psychology, CS, design, business, etc.).
 - Good design must consider **context, user groups, cultural differences, accessibility, UX goals**.
 - **Design principles** (visibility, feedback, constraints, consistency, affordances) help you design and evaluate interfaces.
-

Practice Questions

Now: 5 MCQs, 5 short questions, 1 case study – all from Week 1 only.

A. Multiple-Choice Questions (MCQs)

MCQ 1

Which of the following **best describes Interaction Design** according to Sharp, Rogers, and Preece?

- A. Designing computer hardware that is cheap and efficient
- B. Designing interactive products to support how people communicate and interact in their everyday and working lives
- C. Designing products only for expert users
- D. Designing systems that replace human workers

Correct answer: B

Explanation: Interaction design focuses on **interactive products** (apps, websites, devices) that support **people's communication and interactions** in both everyday and work contexts.

MCQ 2

Which pair correctly matches a **design principle with its description**?

- A. Visibility – restricting possible user actions
- B. Feedback – making different parts of the interface look similar
- C. Constraints – preventing users from performing incorrect actions
- D. Consistency – sending information back to the user after an action

Correct answer: C

Explanation:

- **Constraints** are used to **restrict actions** and prevent errors.
- Visibility = making things obvious.
- Feedback = sending info back.
- Consistency = keeping similar operations similar.

MCQ 3

Which of the following is **NOT** a **usability goal**?

- A. Effective to use
- B. Easy to remember
- C. Enhancing sociability
- D. Safe to use

Correct answer: C

Explanation:

“Enhancing sociability” is more of a **user experience goal** (emotional/social), not a classic usability goal. Usability goals include effectiveness, safety, efficiency, easy to learn, easy to remember, utility, and sometimes enjoyment.

MCQ 4

Which option shows a correct example of **external inconsistency**?

- A. Using different colors for active/inactive buttons in the same app
- B. Using Ctrl+C for copy in one app and Ctrl+X for copy in another
- C. Making the “Save” option greyed out when there are no changes
- D. Hiding advanced settings behind a “More Options” button

Correct answer: B

Explanation:

External consistency means **different applications** use similar rules. Changing shortcut behavior across apps (Ctrl+C vs Ctrl+X for copy) is an external inconsistency.

MCQ 5

Which pair correctly matches the **type of disability** with an example?

- A. Sensory – paralysis of legs
- B. Physical – color blindness
- C. Cognitive – memory loss due to old age
- D. Situational – lifelong use of wheelchair

Correct answer: C

Explanation:

- Cognitive impairment includes memory loss or learning difficulties (e.g., due to old age).
 - Sensory = vision/hearing issues.
 - Physical = movement issues.
 - Situational refers to **temporary environmental** limitations, not lifelong wheelchair use.
-

B. Short-Answer Questions

(Write 3–5 lines for each in the exam.)

SAQ 1

Define accessibility and inclusiveness in interaction design. How are they different?

Model Answer:

Accessibility is about how easily a product can be used by people with **disabilities**, e.g., those who rely on screen readers or voice control. Inclusiveness is broader: it means designing products and services that can be used by as many people as possible, regardless of **disability, age, education, or income**. Accessibility focuses on disabled groups; inclusiveness focuses on *everyone*, including them.

SAQ 2

Explain “pragmatic” and “hedonic” qualities in Hassenzahl’s model of user experience with an example.

Model Answer:

Pragmatic qualities are about **practical usefulness** – how easily a product helps users achieve their goals (e.g., a banking app that lets you transfer money quickly and clearly). Hedonic qualities are about **pleasure and stimulation** – how fun, interesting, or meaningful it feels (e.g., a music app with beautiful animations and playlists that match your mood). Good UX usually balances both.

SAQ 3

List any four usability goals and briefly describe each.

Model Answer:

Four usability goals are:

1. **Effective** – users can successfully complete tasks (e.g., book a ticket).
 2. **Efficient** – tasks can be done quickly with minimum effort.
 3. **Safe** – prevents serious errors, like accidental money transfer.
 4. **Easy to learn** – new users can pick it up without long training.
(You could also mention: good utility, easy to remember, enjoyable to use.)
-

SAQ 4

What is the difference between internal and external consistency? Give one example of each.

Model Answer:

Internal consistency means the **same application** behaves in a similar way throughout – for example, all “Save” buttons in a word processor look and act the same. External consistency means **different applications/devices** follow the same rules – for example, Ctrl+C for copy across many programs. The phone vs calculator keypad number layouts are an example of poor *external* consistency.

SAQ 5

Why is involving users throughout the design process important in interaction design?

Model Answer:

Involving users helps designers understand real needs and workflows instead of relying on assumptions. It reveals differences between user groups (e.g., teenagers vs older adults), surfaces privacy concerns, and uncovers usability problems early. With user feedback and iteration, the final product is usually more usable and delivers a better user experience.

C. Case Study Question

Case Study

You are designing an **online ticketing system** for a large bus service that wants to move from paper tickets and cash counters to a fully digital platform.

The company plans to:

- Let users **search, book, and pay** for tickets via website/app.
- Store user details and previous routes for quick rebooking.
- Show QR codes at the gate instead of physical tickets.

Question:

Using concepts from Week 1, discuss:

1. At least **three design considerations** you must keep in mind about **users and context** (who/what/where).
 2. At least **three usability goals** that are critical for this system.
 3. Two examples of **design principles** (e.g., visibility, feedback, constraints, consistency, affordances) and how you would apply them.
-

Model Answer (Structured)

1. Users and context (who/what/where)

- **Who:**
 - Different ages: students, working adults, and elderly users. Some may be tech-savvy, others not.
 - Some users may have **disabilities**, e.g., low vision or limited mobility → need large fonts, good contrast, screen-reader support.
- **What activities:**
 - Searching routes, comparing timings, selecting seats, entering passenger data, paying online, and retrieving QR tickets later.
- **Where:**
 - On phones while walking, in noisy stations, at home on laptops, or with slow internet connections.
 - So design must be mobile-friendly, clear even in poor lighting, and tolerant of bad connections.

2. Key usability goals

- **Effectiveness:** Users must be able to successfully find and book a bus without confusion.
- **Efficiency:** Booking should take only a few steps; auto-fill previous passenger details to save time.

- **Safety:** Prevent double bookings and payment errors (e.g., confirmation screens, clear refund policies).
- **Easy to learn & remember:** First-time users should understand the flow quickly; returning users should easily repeat bookings.

3. Applying design principles

- **Visibility:**
 - Show clear, large buttons for “Search Buses”, “Select Seat”, “Pay Now”.
 - Display journey summary (date, time, route, seat) before payment so the user sees exactly what they’re buying.
- **Feedback:**
 - After payment, show a clear message (“Payment successful”) and email/SMS confirmation with the QR code.
 - When user clicks “Pay”, show a loading indicator so they know the system is working.
- **Constraints (extra if needed):**
 - Disable the “Confirm Booking” button until all required details are filled.
 - Prevent selecting past dates by greying them out in the calendar.

You could easily extend this answer into a long answer in the exam.